Foundations of Large Language Models

Abilities and Limitations through a Formal Language Learning Perspective

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https://courses.mpi-sws.org/fllm-ws26/

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Large Language Models

A large language model is a parameter-rich neural network trained to model the probability distribution of sequences of words, enabling it to generate, predict, and understand text across diverse contexts.

$$P(s_t \mid s_{t-1}, \ldots, s_1)$$

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Conditional Probability Table: A Faithful Simulation of an LLM

Prefix	Generation	Probability
Germany's capital is	Paris	0.02
	Berlin	0.91
	Washington DC	0.01
	...	...

Vocabulary: $50,000$ tokens
Context length: 2048 tokens
Unique entries $50,000^{2048} \gg 10^{80}$, the number of atoms in the universe

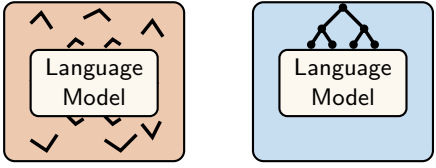
The Central Question

What are the abilities and limitations of LLMs?

The Sciences of the Artificial

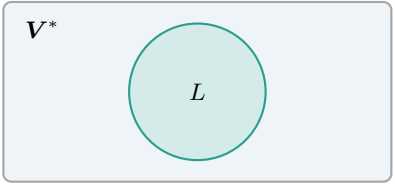
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Formal Language

A formal language is a set (distribution) of strings generated by a formal grammar



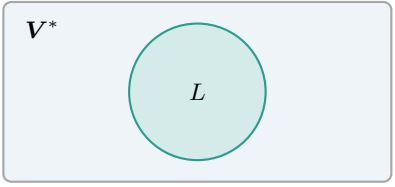
A formal language inside the universe of strings

Formal Language

A formal language is a set (distribution) of strings generated by a formal grammar

$S \to A18\ A16\ [0.50]$ } Level 4
$S \to A16\ A18\ A17\ [0.50]$ } Level 4
$A18 \to A15\ A14\ A13\ [0.50]$
$A18 \to A14\ A15\ A13\ [0.50]$
$A17 \to A14\ A13\ A15\ [0.50]$ } Level 3
$A17 \to A13\ A14\ A15\ [0.50]$ } Level 3
$A16 \to A14\ A15\ [0.50]$ } Level 3
$A16 \to A15\ A14\ [0.50]$ } Level 3
$A15 \to A11\ A12\ A10\ [0.50]$
$A15 \to A12\ A10\ A11\ [0.50]$
$A14 \to A11\ A10\ A12\ [0.50]$ } Level 2
$A14 \to A10\ A11\ A12\ [0.50]$ } Level 2
$A13 \to A10\ A12\ A11\ [0.50]$ } Level 2
$A13 \to A12\ A11\ A10\ [0.50]$ } Level 2
$A12 \to 9\ 8\ 7\ [0.50]$
$A12 \to 8\ 7\ [0.50]$
$A11 \to 6\ 5\ [0.50]$ } Level 1
$A11 \to 6\ 4\ 5\ [0.50]$ } Level 1
$A10 \to 3\ 1\ [0.50]$ } Level 1
$A10 \to 1\ 2\ 3\ [0.50]$ } Level 1

A hierarchical and probabilistic context-free grammar



A formal language inside the universe of strings



A sampled string with its derivation tree

Advantages of Formal Language

- **Precise specification.** We can decide whether a string belongs to the language
- **Exact sampling.** Train and test strings are sampled i.i.d.
- **No semantic ambiguity, no contamination.** Pure syntax, and unseen in pretraining
- **Control over language representation.** The grammar is a knob on language complexity

Our Pursuit, in Three Axes

1

No assumption about the computational model

Which languages are learnable at all?

2

A simplified model of the transformer

What can a transformer express and learn?

3

LLMs in practice

Can we explain the phenomena of LLMs related to language learning?

abstract

concrete

Axis 1: No Assumption about the Computational Model

From positive examples alone, can *any* learner learn a language?

- Can we identify a language from a list of languages?
- Can we generate new strings from the unknown language?

Axis 2: A Simplified Model of the Transformer

What is the *computational model* of a transformer?

- What languages can the transformer express?
- Is expressivity a good proxy for learnability?
- Can we interpret a transformer?

Axis 3: LLMs in Practice

Can we explain the *phenomena* of LLMs related to language learning?

- What languages are difficult for LLMs to learn in practice?
- Which learning modes are most effective?
- What is the implicit world model of an LLM when learning?
- What is the interplay between learning and memorization?

Logistics

Website	https://courses.mpi-sws.org/fllm-ws26
Class	Weekly, Thursday, 2–3:30 pm, Room 112 (MPI-SWS)
Office hours	Weekly, Friday, 2–3:30 pm, Room 112 (MPI-SWS)
Email	fllm26instructors@mpi-sws.org

Grading



- **Presentation.** 45–60 minutes on your paper, then Q&A
- **Report.** Due before the paper is presented, by 2 pm that day
- **Participation.** Engagement in the Q&A

Passing

A combined score of **at least 50%**

What We Expect

A good presentation

- Motivate the research problem, and go deeper into at least one topic
- Teach what you have learned from the paper
- Two presenters: one champion, one critic

A good report on the presented paper

- We give you an LLM-generated review of the paper, and your task is to critique it
- Write what you like about the review and what it misses

Class participation

Read the paper **before** the session and ask questions — the more difficult, the better

Syllabus

Date	Topic	Reading
22 Oct	Short intro and Q&A	—
29 Oct	No class	—
Axis 1 — No assumption about the computational model		
5 Nov	Generation in the Limit	Kleinberg & Mullainathan, *Language Generation in the Limit*, NeurIPS 2024
Axis 2 — A simplified model of the transformer		
12 Nov	Model of a Transformer	Weiss, Goldberg & Yahav, *Thinking Like Transformers*, ICML 2021
19 Nov	Predicting Length Generalization	Zhou et al., *What Algorithms Can Transformers Learn? A Study in Length Generalization*, ICLR 2024
26 Nov	Recovering the Learned Program	Friedman, Wettig & Chen, *Learning Transformer Programs*, NeurIPS 2023
Axis 3 — LLMs in practice		
3 Dec	Difficulty of Learning	Liu et al., *Exposing Attention Glitches with Flip-Flop Language Modeling*, NeurIPS 2023
14 Jan	Implicit World Models	Vafa et al., *Evaluating the World Model Implicit in a Generative Model*, NeurIPS 2024
21 Jan	Learning Modes	Ghosh et al., *Fine-Tuning vs. In-Context Learning in Large Language Models: A Formal Language Learning Perspective*, ACL 2026
28 Jan	Memorization & Learning	Ghosh et al., *Understanding the Interplay between Memorization and Learning in Large Language Models*, NeurIPS 2026

Next Steps



Before you leave today

Select at least three papers you would be happy to present, by 23 October

We assign papers on 26 October

https://courses.mpi-sws.org/fllm-ws26/

Background Reading

Subject background

- Kim, *The Relationship Between the Chomsky Hierarchy and Automata*, 2019
  math.uchicago.edu/~may/REU2019/REUPapers/Kim.pdf
- Alammar, *The Illustrated Transformer*, 2018
  jalammar.github.io/illustrated-transformer

For your presentation and report

- Ernst, *How to Give a Technical Talk*
  homes.cs.washington.edu/~mernst/advice/giving-talk.html
- Peyton Jones, Hughes & Launchbury, *How to Give a Good Research Talk*, 1993
  microsoft.com/.../giving-a-talk.pdf
- Smith, *The Task of the Referee*, 1990
  pages.cs.wisc.edu/~markhill/the_task_of_the_referee.pdf